

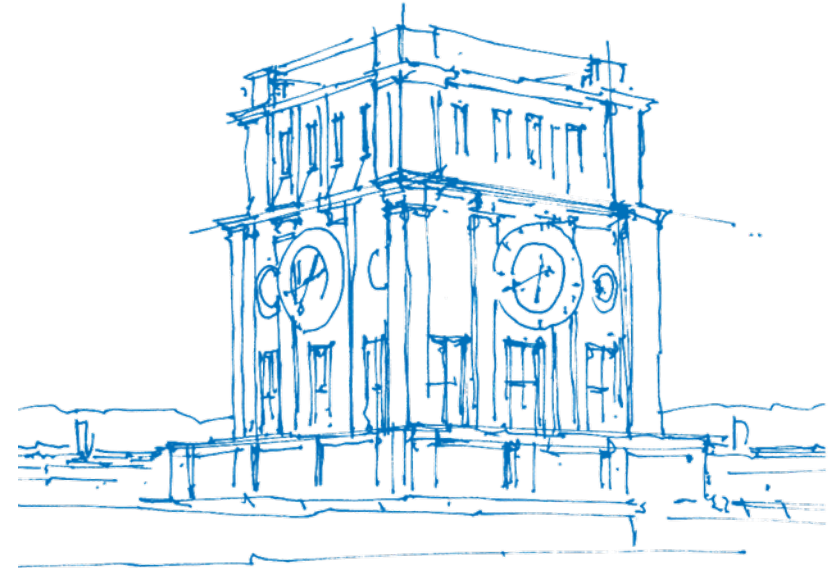
Cost-Aware Optimized Front-Door Experimental Design

Leopold Mareis

Department of Mathematics

Technical University of Munich (TUM)

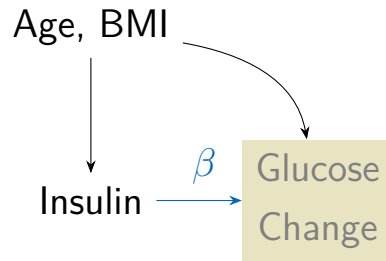
(joint work with Mathias Drton)



TUM Uhrenturm

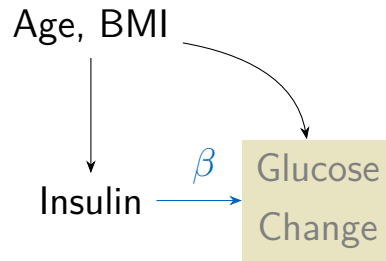
Selective Measurement in Clinical Trials

Example: Effect β of insulin on blood glucose in ICU patients (MIMIC-IV, $n = 999$).



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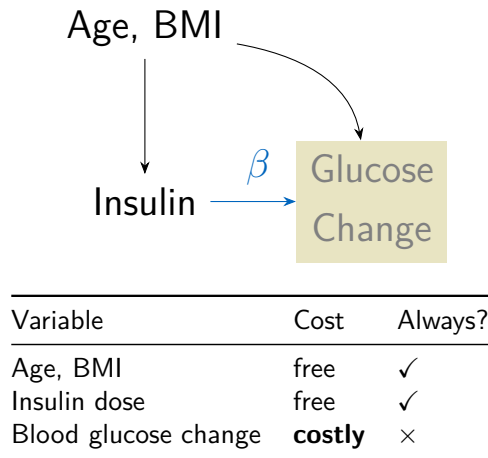
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Variable	Cost	Always?
Age, BMI	free	✓
Insulin dose	free	✓
Blood glucose change	costly	×

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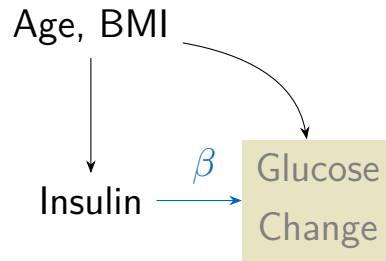
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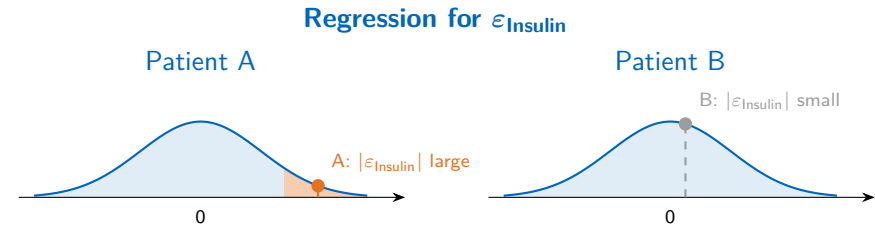
Inefficiency: Follow-up for all patients is expensive.

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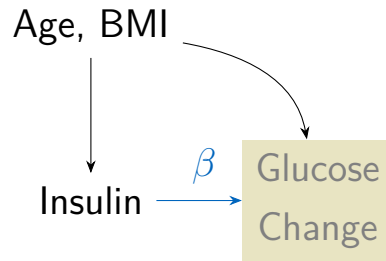
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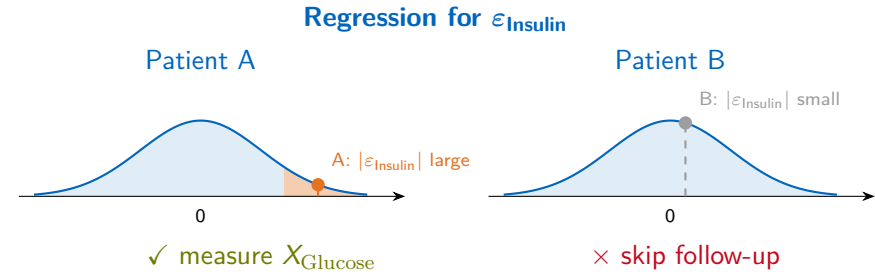
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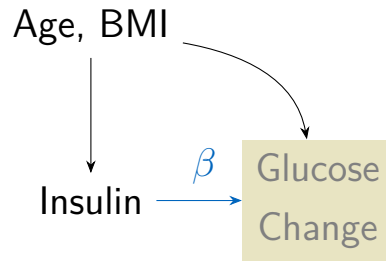
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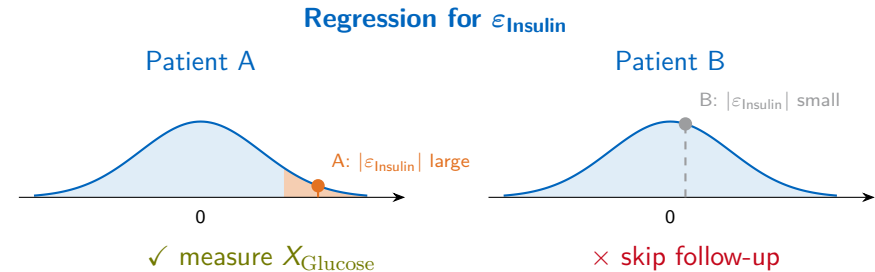
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⇒ Optimize sequential measurement decisions driven by propensity $\pi(X)$.

Inefficiency: Follow-up for all patients is expensive.

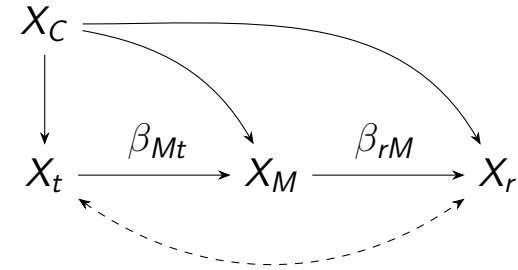
Partial Observation Model $\mathcal{M}_\pi$

Goal: Estimate the causal effect $\xi = \beta_{Mt} \beta_{rM}$ of a treatment X_t on a response X_r .

SEM: Full data $X = (X_C, X_t, X_M, X_r)$ solves

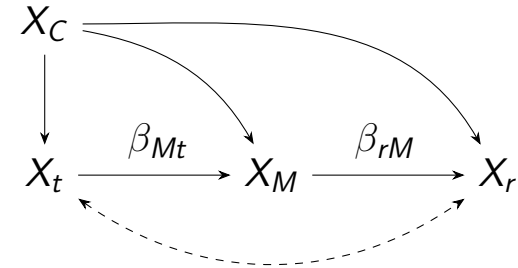
$$\begin{pmatrix} X_C \\ X_t \\ X_M \\ X_r \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ \beta_{tC} & 0 & 0 & 0 \\ \beta_{MC} & \beta_{Mt} & 0 & 0 \\ \beta_{rC} & 0 & \beta_{rM} & 0 \end{pmatrix} \begin{pmatrix} X_C \\ X_t \\ X_M \\ X_r \end{pmatrix} + \begin{pmatrix} \varepsilon_C \\ \varepsilon_t \\ \varepsilon_M \\ \varepsilon_r \end{pmatrix},$$

$$\mathcal{P}_\varepsilon = \mathcal{P}_{\varepsilon_C} \otimes \mathcal{P}_{(\varepsilon_t, \varepsilon_r)} \otimes \mathcal{P}_{\varepsilon_M}.$$



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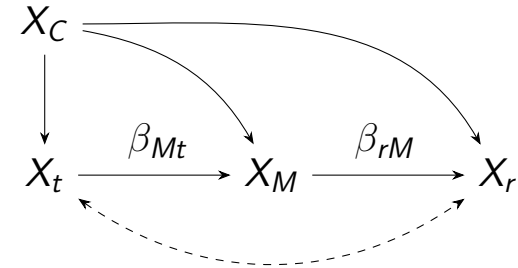


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Δ	X_C	X_t	X_M	X_r
2	0.4	1.3	-0.7	.
∞	1.2	0.8	1.5	-0.3
1	-1.1	0.2	.	.
2	1.4	-0.9	0.6	.



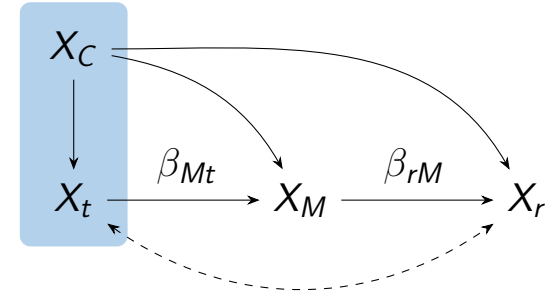
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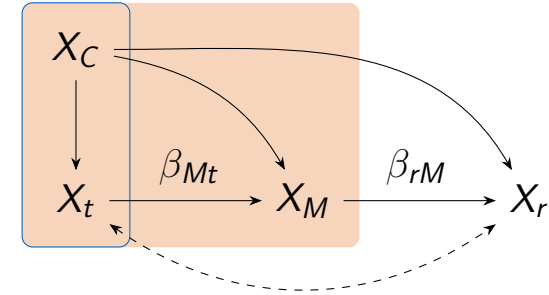
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Measure X_M with probability $\pi_1(X_{Ct})$

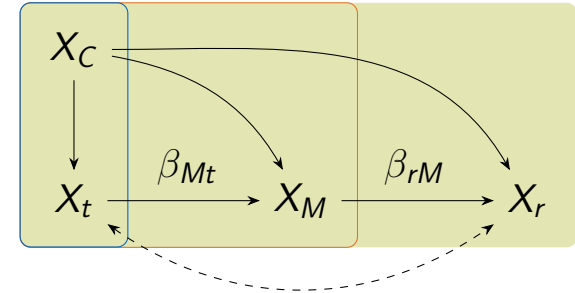
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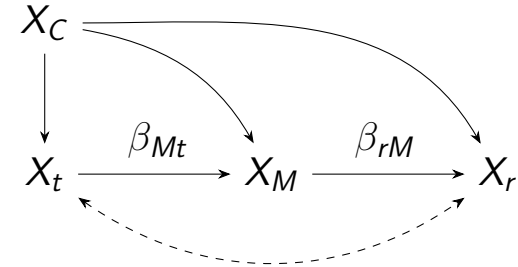


Measure X_r with probability $\pi_2(X_{CtM})$

Δ	$G_\Delta(X)$	cost
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∞	X_{CtMr}	$c_0 + c_1 + c_2$

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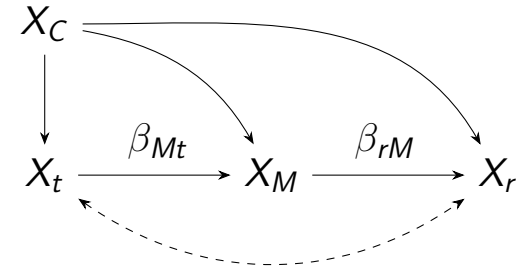


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- **Model class $\mathcal{M}_\pi$:** Observed data distributions of $(\Delta, G_\Delta(X))$ governed by $\pi = (\pi_1, \pi_2)$.
- **Full-data model $\mathcal{M}_1$:** $\pi \equiv (1, 1)$, so $G_\Delta(X) \equiv X_{CtMr}$.

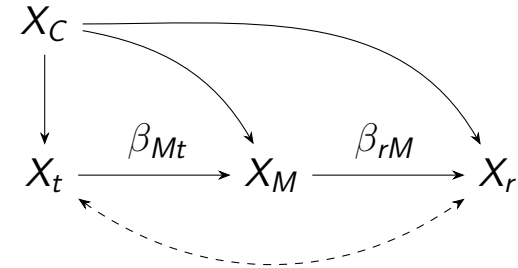


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Problem:

$$\min_{\pi=(\pi_1, \pi_2), \text{RAL } \hat{\xi}_n} \text{Var}_\infty(\hat{\xi}_n; \pi), \text{ s.t.}$$

$$\mathbb{E}[c_0 + \pi_1(X_{Ct}) c_1(X_{Ct}) + \pi_1(X_{Ct}) \pi_2(X_{CtM}) c_2(X_{CtM})] = b_0.$$

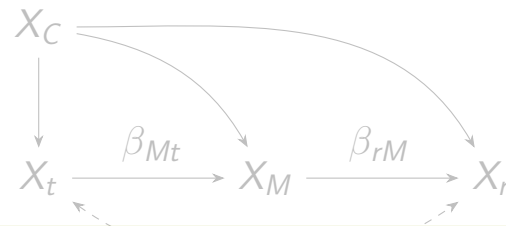
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π_1

π_2

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- **Model class $\mathcal{M}_\pi$:** Observed data distributions of

(Δ)

Two mathematical challenges:

- **Full**
 1. **Estimators $\hat{\xi}_n$:** Influence functions φ characterize estimators on $\mathcal{M}_\pi$ and yield $\text{Var}(\hat{\xi}_n; \pi)$.
 2. **Optimized design π^* :** Solve the variational problem $\min_\pi \text{Var}_\infty(\hat{\xi}_n; \pi)$.

Problem:

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RAL Estimators and Influence Functions

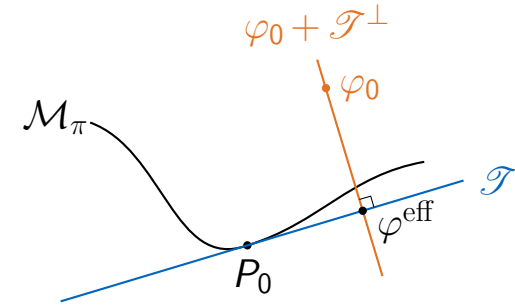
$\hat{\xi}_n$ is **regular and asymptotically linear (RAL)** for ξ if

$$\sqrt{n}(\hat{\xi}_n - \xi) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \varphi(X^{(i)}) + o_{\mathcal{P}}(1)$$

for some **influence function** φ , so that

$$\sqrt{n}(\hat{\xi}_n - \xi) \xrightarrow{d} \mathcal{N}(0, E[\varphi\varphi^\top]).$$

Goal: $\operatorname{argmin}_{\varphi} E[\varphi\varphi^\top]$



RAL Estimators and Influence Functions

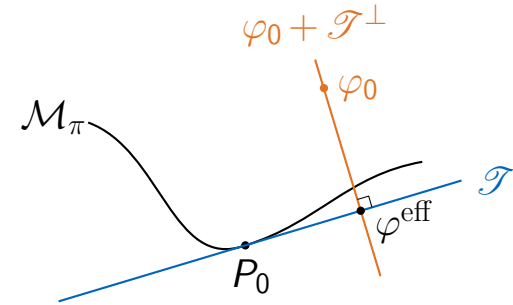
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Goal: $\arg\min_{\varphi} E[\varphi\varphi^\top] = \varphi^{\text{eff}} = \Pi(\varphi_0 | \mathcal{T})$ yields minimal asymptotic variance $\hat{\xi}_n(\varphi^{\text{eff}})$.



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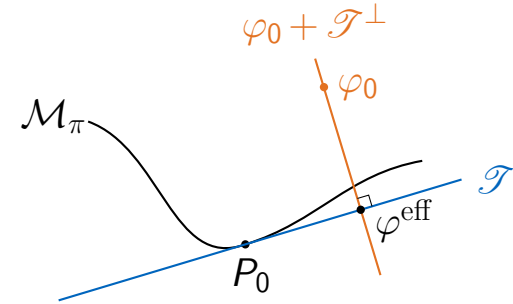
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Insight [Tsiatis, 2006]: Find $\varphi_{\xi}^{\text{eff}}$ the full-data model $\mathcal{M}_1$; augment to the partial-data models $\mathcal{M}_{\pi}$.

Estimator $\hat{\xi}_n$ in $\mathcal{M}_\pi$

Theorem 1. Efficient influence function for $\xi = \beta_{rM}\beta_{Mt}$ in $\mathcal{M}_1$:

$$\varphi_\xi^{\text{eff}} = \beta_{rM} \varphi_{\beta_{Mt}}^{\text{eff}} + \varphi_{\beta_{rM}}^{\text{eff}} \beta_{Mt},$$

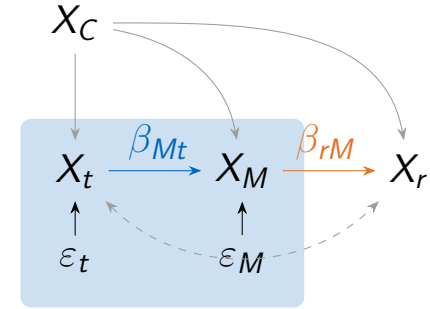
$$\varphi_{\beta_{Mt}}^{\text{eff}} = \varepsilon_M \varepsilon_t \text{Var}(\varepsilon_t)^{-1}, \quad \varphi_{\beta_{rM}}^{\text{eff}} = \varepsilon_r^\perp \varepsilon_M^\top \text{Var}(\varepsilon_M)^{-1}.$$

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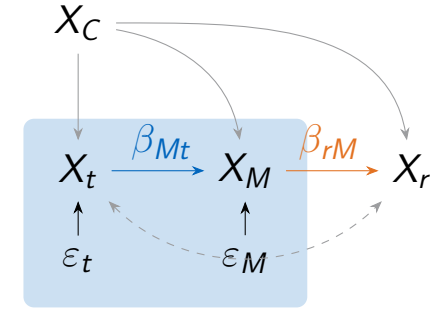
Estimator: $\hat{\beta}_{Mt,n}$ s.t. $E[\hat{\varepsilon}_t \hat{\varepsilon}_M] \stackrel{!}{=} 0$

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Optimal augmentation of φ_ξ^{eff} in $\mathcal{M}_\pi$ [Tsiatis, 2006]:

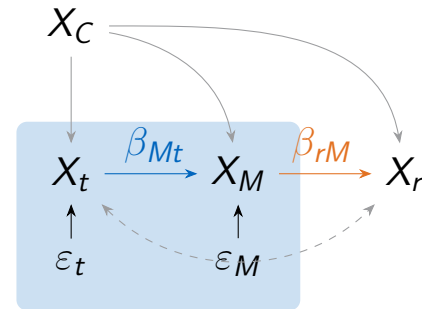
$$\varphi_\xi^{\text{opt},\pi} = \frac{I(\Delta \geq 2)}{\pi_1(X_{Ct})} \beta_{rM} \varphi_{\beta_{Mt}}^{\text{eff}} + \frac{I(\Delta = \infty)}{\pi_1(X_{Ct})\pi_2(X_{CtM})} \varphi_{\beta_{rM}}^{\text{eff}} \beta_{Mt}.$$

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Asymptotic Variance:

$$\text{Var}_\infty(\hat{\xi}_n; \pi) = E \left[\frac{g_1(X_{Ct})}{\pi_1(X_{Ct})} \right] + E \left[\frac{g_2(X_{CtM})}{\pi_1(X_{Ct})\pi_2(X_{CtM})} \right], \quad \left(\begin{array}{l} g_1(X_{Ct}) := \beta_{rM} \text{Var}(\varepsilon_M) \beta_{rM}^\top \frac{\varepsilon_t^2}{\text{Var}(\varepsilon_t)^2} \\ g_2(X_{CtM}) := \text{Var}(\varepsilon_r | \varepsilon_t) (\varepsilon_M^\top \text{Var}(\varepsilon_M)^{-1} \beta_{Mt})^2 \end{array} \right)$$

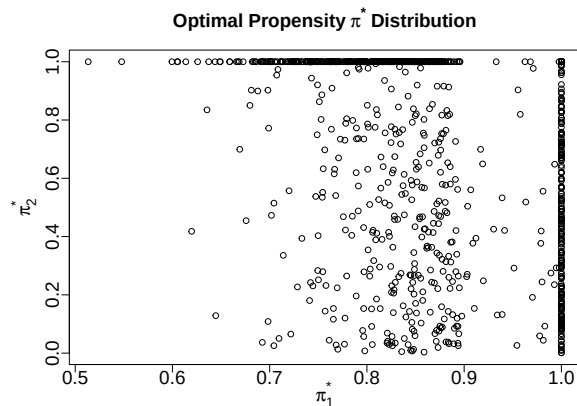
Optimized Design π^*

Theorem 2. $\hat{\xi}_n^{\text{opt}}$ in $\mathcal{M}_{\pi^*}$ minimizes estimation variance¹ of ξ under budget constraint ($\lambda > 0$ multiplier):

$$\pi_1^* = \min \left(1, \max \left(\sqrt{\frac{g_1}{\lambda c_1}}, \sqrt{\frac{g_1 + \mathbb{E}[g_2 | \varepsilon_{C,t}]}{\lambda (c_1 + \mathbb{E}[c_2 | \varepsilon_{C,t}])}} \right) \right), \quad \pi_2^* = \min \left(1, \sqrt{\frac{g_2 c_1}{g_1 c_2}} \right).$$

Intuition: Measure further proportionally to the information-to-cost ratio $\sqrt{g_j/c_j}$:

$$g_1(X_{Ct}) = \underbrace{\frac{\beta_{rM} \text{Var}(\varepsilon_M) \beta_{rM}^\top}{\text{Var}(\varepsilon_t)^2}}_{\text{constant}} \cdot \underbrace{\varepsilon_t^2}_{\text{leverage for } \hat{\beta}_{Mt}}.$$



¹Over augmentations of $\varphi^{F,\text{eff}}$; see arxiv.org/pdf/2603.22024 for details.

Applications and Code

Application	n	Rel. efficiency
Sachs	853	76.6%
Sachs	853	68.0%
MIMIC-III (ICU)	999	90.0%
causalAssembly	3,000	94.7%

Rel. efficiency = $\text{Var}_{\infty}(\hat{\xi}_n; \pi^*) / \text{Var}_{\infty}(\hat{\xi}_n; 1)$

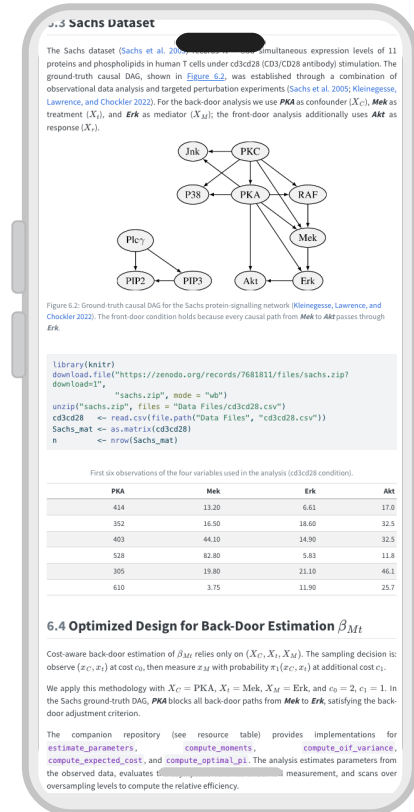
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<https://st-mardi.quarto.pub/gmci/>

Conclusion

- Efficient influence function for the front-door effect $\xi = \beta_{rM}\beta_{Mt}$ in $\mathcal{M}_1$, and its optimal augmentation.
- Closed-form sequential design π^* based on unit-level information-to-cost ratios g_j/c_j .
- Optimized **back-door** experimental design follows as a corollary.
- 5–32% variance reduction over full measurement in biological, medical, and industrial datasets.

Outlook: Efficient estimation in general ADMGs. Sampling designs that depart from the causal ordering.

- [A. Tsiatis.](#)
Semiparametric Theory and Missing Data.
Springer Series in Statistics. Springer, New York, 2006.
- [L. Mareis and M. Drton.](#)
Cost-Aware Optimized Front-Door Experimental Design.
To appear: *Proceedings of Machine Learning Research*, PMLR, 2026. Preprint:
[arXiv:2603.22024](#).



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